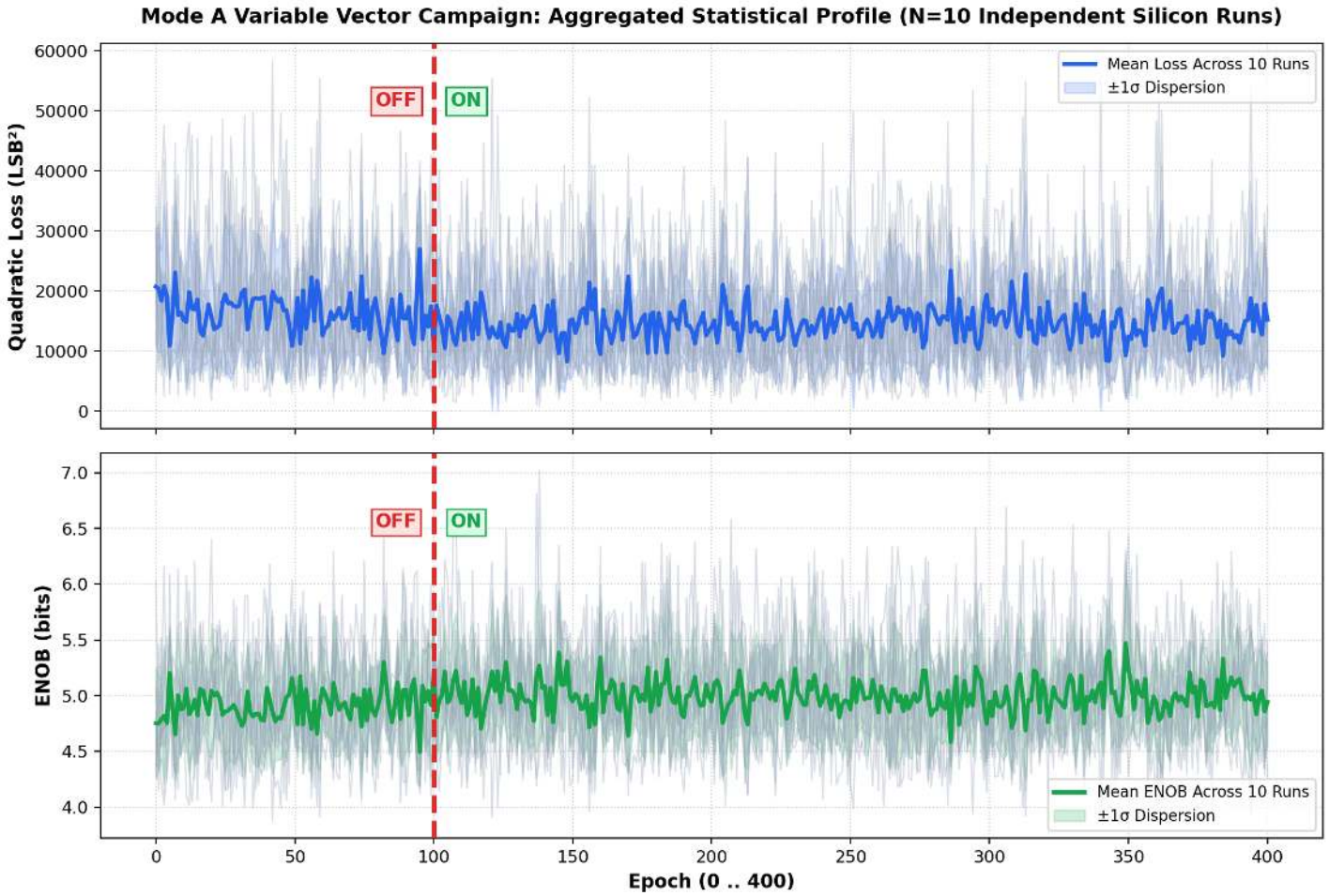
Inverted MAML Mode A: 10 Variable Vectors Statistical Report

Empirical Multi-Vector Silicon Benchmark | Continuous Rotating In-Data

Executive Summary: This empirical silicon study presents statistical characterization of Inverted MAML Mode A hardware on the Efinix Trion T120 Master SoC. The experiment protocol strictly evaluates **Phase 1: Baseline OFF (Epochs 0..99)** with training disabled (LR=0.0, weights locked at physical null W8=24, W3=64) transitioning to **Phase 2: Training ON (Epochs 100..400)** with active 2-layer adaptation (LR=0.25x). Tested across 10 runs with continuously rotating random vectors (new vector every epoch).

Phase 1: Baseline OFF (Ep 0..99)	Phase 2: Training ON (Ep 100..400)	Peak Silicon Control
Mean Loss: **16528.9 ± 1515.8 LSB²** Mean ENOB: **4.91 ± 0.07 b**	Mean Loss: **14554.7 ± 464.6 LSB²** Mean ENOB: **5.00 ± 0.03 b**	Best Loss: **748.6 LSB²** Best ENOB: **7.02 b**

Aggregated Statistical Convergence Profile



Individual Run Breakdown (10 Runs × 400 Epochs)

Run #	Input Vector X (Cols 0..5)	Base Loss (LSB²)	Train Loss (LSB²)	Final ENOB	Best Ep / ENOB	Drop %
Run 01	[-25, -40, -34, -23, -34,...	15669 ± 9615	13963 ± 7553	**5.22 b**	#126 (6.50 b)	**88.3%**
Run 02	[-27, 42, 3, 44, 41, 23]	16114 ± 9689	14537 ± 8450	**4.26 b**	#265 (6.34 b)	**93.4%**
Run 03	[12, 0, -12, 21, 12, 18]	14294 ± 7994	14126 ± 7886	**4.95 b**	#349 (6.30 b)	**32.2%**
Run 04	[-24, 1, 0, -26, 37, -10]	15578 ± 9720	14328 ± 8291	**5.50 b**	#343 (6.49 b)	**91.1%**
Run 05	[-32, -32, 36, -10, 28, 35]	15615 ± 9364	14850 ± 8653	**4.84 b**	#207 (6.58 b)	**95.5%**
Run 06	[-13, 26, 44, -3, -6, 16]	17136 ± 10149	14257 ± 8077	**4.79 b**	#306 (6.69 b)	**89.3%**
Run 07	[-40, 0, 27, -37, -4, -10]	16145 ± 8060	14026 ± 8584	**5.22 b**	#137 (6.81 b)	**96.0%**
Run 08	[-31, 41, 5, 19, 12, -17]	16833 ± 8219	14983 ± 8056	**5.38 b**	#138 (7.02 b)	**95.0%**
Run 09	[-3, 25, -45, -31, 13, 8]	20172 ± 10280	15237 ± 8648	**4.74 b**	#194 (6.38 b)	**94.6%**
Run 10	[-15, -40, 34, -25, 8, -17]	17733 ± 11313	15239 ± 8469	**4.52 b**	#237 (6.30 b)	**92.9%**